

ERNESTO BONILLA-HOSHIKAWA

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OBJECTIVE

Seeking a Fall 2026 co-op in embedded systems, robotics, AI, or software engineering, applying experience with FPGA development, C++, hardware-software integration, and backend systems in hands-on engineering environments.

EDUCATION

Northeastern University, Boston, MA

B.S. in Computer Engineering & Computer Science (GPA: 3.97)

Expected May 2028

Honors: Merit Scholarship Recipient; Tau Beta Pi Engineering Honor Society.

Relevant Coursework: Embedded Systems, Circuits & Signals, OOP, Fundamentals of Networks, Logic & Computation, Discrete Math, Differential Equations, Physics: Mechanics & EM.

Saint Peter's Preparatory School, Jersey City, NJ

September 2020 – June 2024

Honors: Magna Cum Laude (GPA: 4.42), Ignatian Scholar (Merit Scholarship).

TECHNICAL SKILLS

- **Languages:** C++, Python (pandas), Java, JavaScript, Bash, SQL.
- **Embedded/FPGA (DE1-SoC):** Cyclone V (Intel/Altera), Quartus Prime Lite, RTL/top-level integration, HPS–FPGA integration, AXI, memory-mapped I/O (mmap), register-level control, FSMs, timers/counters, PWM.
- **Embedded Linux & Tools:** Linux user-space peripheral control, debugging, Git.
- **Hardware/Test:** ADALM2000 (oscilloscope/logic analyzer), NI USB-DAQ, LTspice, breadboarding/debug, AD627 instrumentation amplifier, Arduino, I2C, SPI.
- **Backend/Cloud (Secondary):** Node.js, Express, REST APIs, MySQL; AWS (S3, RDS, Redshift).
- **Certifications:** AWS Certified Cloud Practitioner; AWS Certified AI Practitioner.

PROFESSIONAL EXPERIENCE

Khoury College of Computer Sciences, Boston, MA — Teaching Assistant — Logic & Computation *January 2026 – Current*

- Hold office hours to assist students with formal logic, algorithmic reasoning, and proof techniques.
- Communicate complex technical concepts to individuals and small groups to facilitate learning.

Optimus Health Analytics, New York, NY — Data Engineer Intern

June – August 2023

- Automated ETL workflows on ~1M dataset using Bash and cron on Linux servers, reducing processing time by ~20%.
- Reduced ingestion failures ~30% by scripting data validation and type normalization (missing fields, invalid rows).

TECHNICAL PROJECTS

Programmable Spider Robot | C++, Embedded Linux, FPGA Cyclone V

December 2025

- Developed gait-control logic in C++ using OOP to manage hexapod movement behaviors on a DE1-SoC.
- Iteratively tested and calibrated gait sequences by monitoring terminal output and observing robot behavior in real time.
- Diagnosed and resolved startup failures caused by timing delays at initialization, enabling consistent robot operation.

Custom ECG System | LTspice, NI USB-DAQ, MATLAB

November - December 2025

- Designed an analog front end using an AD627 amplifier with a gain of ~775.
- Implemented hardware filters (Analog Bandpass 0.1Hz–250Hz) and digital filters (Notch, Park-McClellan) to assess signal quality.
- Captured heart rate data (70 BPM) and analyzed signals in MATLAB to assess noise performance.

FPGA Motor Control Lab | Quartus Prime Lite, C++, Embedded Linux, FPGA Cyclone V

November 2025

- Implemented PWM generation for RC Servos (50MHz clock, 20ms period) and DC motors.
- Developed a finite-state machine (FSM) for stepper-motor control with parameterized step rates.
- Built a Linux user-space controller using mmap and the AXI bus to write memory-mapped registers.
- Validated timing behavior and duty cycles using oscilloscope measurements against expected simulation results.

Stand AI (Oasis Program) | Node.js, Express, OpenAI API | [GitHub](#)

January – March 2025

- Built a backend API for an LLM fact-checking tool producing labels, confidence scores, and citations.
- Collaborated in an agile-style team environment focused on experiential learning and project-based CS.

ACTIVITIES & HONORS

- AI @ Northeastern — AI Applications Workshop (prompt engineering, REST API).
- Oasis Program — experiential learning, teamwork, mentorship, project-based CS.